

Estimation

ECON 97: Economic Toolkit

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<https://join.iclicker.com/RDNG>

Quiz: continuous variables

Let X have PDF:

$$f(x) = cx^3 \quad x \in [0, 2]$$

What is $P(0 < X < 1)$?

- A. 1/16
- B. 7/16
- C. 5/16
- D. 3/16

Quiz: uniform distributions

Suppose that

$$X \sim U[0, b], \quad b > 1$$

$$P\left(\frac{1}{3} < X < \frac{2}{3}\right) = \frac{1}{9}$$

What is $\text{Var}(X)$?

A. $1/2$

B. 1

C. $3/4$

D. $1/4$

U-substitution practice

Suppose X has PDF:

$$f(x) = cx(1 + x^2)^2, \quad 0 \leq x \leq 1.$$

Find c .

U-substitution practice

Suppose X has PDF:

$$f(x) = cxe^{-x^2}, \quad x \geq 0.$$

Find c .

U-substitution practice

Suppose X has PDF:

$$f(x) = \frac{3}{8}(x+1)^2, \quad -1 \leq x \leq 1.$$

Find

$$P(0 \leq X \leq 1).$$

Integrating absolute value functions

Suppose you have:

$$\int_{-2}^3 |x| dx$$

First find where the inside equals zero:

$$x = 0$$

Rewrite the absolute value piecewise:

$$|x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

Split the integral at $x = 0$:

$$\begin{aligned} \int_{-2}^3 |x| dx &= \int_{-2}^0 (-x) dx + \int_0^3 x dx \\ &= \left[-\frac{x^2}{2} \right]_{-2}^0 + \left[\frac{x^2}{2} \right]_0^3 = \frac{13}{2} \end{aligned}$$

Absolute value practice

Suppose X has PDF:

$$f(x) = c|x - 2|, \quad 0 \leq x \leq 4.$$

Find c .

Standard Normal Distributions (Z)

If a variable is normally distributed with $\mu = 0$ and $\sigma^2 = 1$

$$Z \sim N(0, 1)$$

$$F(x) = \Phi(x) = P(Z \leq x)$$

$$\Phi(-x) = 1 - \Phi(x)$$

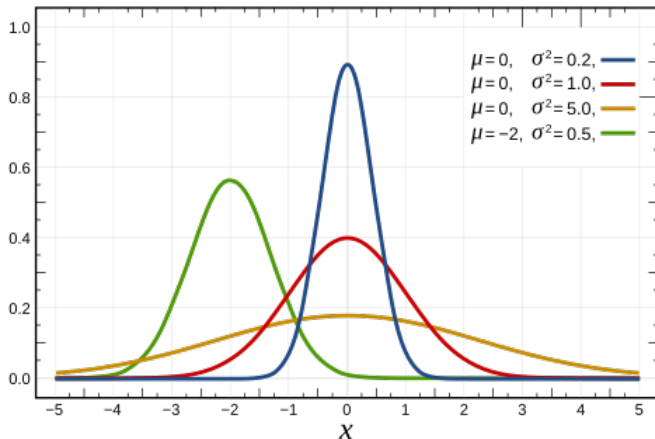
$$\Phi(0) = 1/2$$

$$z_\alpha \implies \Phi(z_\alpha) = 1 - \alpha \text{ (percentile)}$$

If $X \sim N(\mu, \sigma^2)$:

$$\frac{X - \mu}{\sqrt{\sigma^2}} = Z \quad (\text{z-score})$$

Normal vs. Standard Normal visualized



* Standard normal is in red

Normal distribution practice

If $X \sim N(7, 25)$, find:

- $P(X \leq 0.45)$
- $P(|X - 7| \leq 10)$

Find the values of:

- $z_{0.1}$
- $-z_{0.05}$

Normal distribution practice

If $X \sim N(7, 25)$, what is the smallest x among the four choices below such that $P(X > x) = 0.05$

- A. 13.41
- B. 15.225
- C. 16.8
- D. 19.88

Sample Mean and Sample Variance

Suppose we observe a sample of data

$$x_1, x_2, \dots, x_n.$$

Sample Mean: average value of the data

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$$

Sample Variance: how spread out the data are around the mean

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Sample Statistics as “estimators”

Population

$$\mu = E(X)$$

$$\sigma^2 = \text{Var}(X)$$

Sample

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{estimates} \quad \mu$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{estimates} \quad \sigma^2$$

* As n increases, estimators tend to get closer to the “true” population values

Sample mean as R.V.

The sample mean changes every time we observe new data

→ Therefore, $\bar{X}$ is a random variable

$$E[\bar{X}] = \mu$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

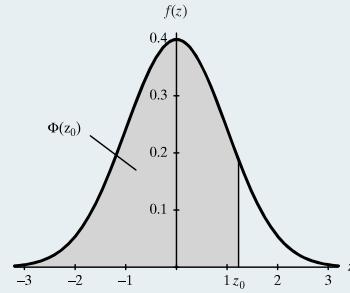
Sample mean & variance practice

Let $\bar{X}$ be the mean of a random sample of size 9 from distribution:

$$f(x) = x/6, \quad x = 1, 2, 3$$

Find the mean and variance of $\bar{X}$.

Table Va: The Standard Normal Distribution Function



$$P(Z \leq z) = \Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-w^2/2} dw$$

$$\Phi(-z) = 1 - \Phi(z)$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
α	0.400	0.300	0.200	0.100	0.050	0.025	0.020	0.010	0.005	0.001
z_α	0.253	0.524	0.842	1.282	1.645	1.960	2.054	2.326	2.576	3.090
$z_{\alpha/2}$	0.842	1.036	1.282	1.645	1.960	2.240	2.326	2.576	2.807	3.291